AI & EMERGING TECHNOLOGY

AI & LLM Security Assessment

1,005+ attack prompts across all 10 OWASP LLM categories — plus MCP testing and adversarial red teaming.

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KEY BENEFITS

Why This Matters

1,005+ LLM Attack Prompts
The most comprehensive AI adversarial testing suite — covering all 10 OWASP LLM categories.

MCP Testing — Industry First
Testing Model Context Protocol implementations before most vendors know they exist.

Exhaustive Escalation
We don't stop at first bypass — escalation until complete compromise or confirmed resilience.

YOUR CHALLENGE

Your organization deploys AI/ML systems, LLMs, and agentic AI tools introducing entirely new attack surfaces:

- Prompt injection and jailbreak vulnerabilities
- MCP tool chains creating privilege escalation vectors
- Training data poisoning and model supply chain risks
- Regulatory pressure from NIST AI RMF and EU AI Act

Why Technieum as Your Trusted Partner
Technieum brings 6 proprietary AI platforms, an in-house R&D team actively hunting zero-days, and AI-powered methodologies that deliver validated, exploitable findings with guaranteed zero false positives — enabling your team to focus remediation effort where it matters most.

OUR SOLUTION

Services Overview

Technieum's AI security practice combines our LLM Attack Suite, MCP testing, and adversarial red teaming — aligned to OWASP LLM Top 10, NIST AI RMF, and MITRE ATLAS.

1	**AI & LLM Penetration Testing**	Comprehensive adversarial testing using Technieum's LLM Attack Suite — 1,005+ attack prompts across all 10 OWASP LLM Top 10 categories. Tests for direct and indirect prompt injection, jailbreak attacks, training data extraction, PII leakage, insecure output handling, excessive agency exploitation, and model denial of service with exhaustive escalation until bypass or confirmed resilience.
2	**MCP Security Testing**	Evaluates Model Context Protocol server implementations — testing for tool injection attacks that manipulate AI agent behavior, privilege escalation through MCP tool chains, data exfiltration via MCP tool access, resource manipulation, authentication and authorization weaknesses in the MCP communication layer, and rug-pull scenarios where trusted tools become malicious.
3	**AI Systems Audit**	NIST AI RMF-aligned audit — governance, data pipelines, model training, deployment security.
4	**Adversarial Red Teaming**	Multi-modal attacks across text, image, audio, and API inputs testing real model robustness.
5	**AI Threat Modelling**	Maps AI attack surfaces — training data poisoning, model supply chain, tool-use chain vulns.
6	**Security Tooling Guidance**	Guardrail implementation — input validation, output filtering, prompt injection detection.

THE TECHNIEUM EDGE

Why Technieum for AI Security

Our Methodology
LLM Attack Suite deployment → MCP testing → adversarial red teaming with escalation → NIST AI RMF-aligned risk assessment → remediation guidance.

LLM Attack Suite
1,005+ prompts, automated simulation, real-time scoring, remediation reports.

MCP Testing
First-mover testing of Model Context Protocol server implementations.

NIST AI RMF Alignment
Governance-ready output mapped to Govern-Map-Measure-Manage.

Adversarial Red Teaming
Multi-modal attacks testing real robustness, not just basic injection.

Exhaustive Escalation
Every attack vector exhausted until compromise or confirmed resilience.

R&D-Driven Intelligence
Dedicated researchers tracking emerging AI attack techniques.

INDUSTRY-SPECIFIC TESTING

Vertical-Specific AI Security

Technieum delivers purpose-built AI security test scenarios tailored to the unique risks, regulations, and threat models facing AI adoption in each industry vertical. Our LLM Attack Suite includes industry-specific prompt libraries, compliance-mapped test cases, and vertical-aware risk scoring.

BY INDUSTRY

Vertical-Specific AI Security Testing

Purpose-built test scenarios for the unique risks facing AI adoption in each vertical

Banking
Financial Services

- MNPI leakage testing
- Unauthorized advice detection
- AML/KYC compliance gaps
- Fiduciary duty breaches
- Reg BI suitability testing
- Proprietary strategy exposure

Telecom
Communications

- Voice AI hijacking
- Subscriber data exposure
- SIM swap facilitation
- Call routing manipulation
- Network config disclosure
- PII extraction via voice AI

Real Estate
Property

- Fair housing violations
- Steering by protected class
- Lending bias detection
- Valuation discrimination
- Advertising exclusion
- Source of income rejection

E-Commerce
Retail

- Price manipulation
- Fake review generation
- Payment fraud facilitation
- Recommendation bias
- Competitor data extraction
- Personalization exploitation

Healthcare
Medical

- PHI exposure testing
- Misdiagnosis risk assessment
- Prescription error scenarios
- Clinical data leakage
- HIPAA compliance gaps
- Telehealth bot security

Teen Safety
Child Protection

- Predator grooming patterns
- Age verification bypass
- Harmful content generation
- COPPA violations
- Parental control bypass
- Inappropriate relationship sim

Airlines
Aviation

- Safety protocol bypass
- Booking manipulation
- Passenger data theft
- Crew schedule interference
- Loyalty program fraud
- Maintenance data exposure

Banking & Financial Services

MNPI (Material Non-Public Information) leakage testing, unauthorized investment advice detection, AML/KYC compliance gap identification, proprietary trading strategy exposure, market manipulation facilitation testing, and fiduciary duty breach scenarios. Aligned with FINRA Rule 3110, SEC Regulation BI, and SR 11-7 model risk management requirements.

Telecommunications

Voice AI agent hijacking, subscriber data exposure through conversational AI, SIM swap facilitation testing, call routing manipulation, network configuration disclosure, and customer PII extraction via voice/text AI assistants. Tests both inbound customer service AI and outbound AI-powered communications.

Real Estate

Fair housing compliance testing across all 7 federally protected classes — race, color, religion, national origin, sex, familial status, and disability. Tests for AI-driven steering, discriminatory listing language generation, lending bias in pre-qualification bots, valuation bias in automated models, and advertising exclusion. Aligned with Fair Housing Act, ECOA, and HUD guidance.

E-Commerce

Dynamic pricing manipulation, fake review generation and detection bypass, payment fraud facilitation, inventory manipulation, competitor intelligence extraction, personalization algorithm exploitation, and product recommendation bias. Tests AI-powered search, recommendation, pricing, and customer service systems.

Healthcare & Medical

PHI (Protected Health Information) exposure through clinical AI assistants, misdiagnosis risk from AI-assisted diagnostic tools, prescription error scenarios, patient data extraction via conversational interfaces, clinical trial data

leakage, and HIPAA compliance gap identification. Tests EHR-integrated AI, telehealth bots, and clinical decision support systems.

Teen Safety & Child Protection

Predator grooming pattern facilitation testing, age verification bypass scenarios, harmful content generation (self-harm, substance abuse, violence), inappropriate relationship simulation, child data collection compliance (COPPA), and parental control circumvention. Critical for any AI platform accessible to minors including social media, gaming, education, and messaging applications.

Airlines & Aviation

Safety protocol bypass testing in AI-assisted flight operations, booking system manipulation, loyalty program fraud facilitation, passenger data extraction, crew scheduling interference, and maintenance procedure circumvention through AI assistants. Tests customer-facing booking AI, operational support systems, and crew communication tools.

Standards Alignment

Every Technieum engagement aligns with globally recognized security frameworks including OWASP Top 10, OWASP API Top 10, OWASP LLM Top 10, PTES, NIST CSF, NIST AI RMF, MITRE ATT&CK, CIS Benchmarks, IEC 62443, and ISO 27001:2022. Our team holds OSCP, OSEP, CRTE, CRTP, eWPTX, eCPPT, and CEH certifications. Technieum is ISO 9001:2015 and ISO 27001:2022 certified.

OUR METHODOLOGY

How We Deliver Results

Technieum's AI security methodology follows a structured five-phase approach purpose-built for LLM applications, AI agents, MCP implementations, and machine learning systems — aligned to OWASP LLM Top 10 and NIST AI RMF.

Discover	Analyze	Attack	Report	Remediate
Asset enumeration	ToIP intelligence	Manual exploitation	CVSS/CWE scoring	AI-powered fixes
Tech fingerprinting	Logic extraction	Business logic test	PoC evidence	Stack-specific code
OSINT gathering	Threat modelling	Chain validation	Exec summary	Config guidance
Scope validation	Risk prioritization	Privilege escalation	Test case docs	Retest validation
Port scanning	Attack planning	Lateral movement	Playbook docs	Closure sign-off
Service detection	Playbook generation	Data exfiltration	Risk heatmap	Knowledge transfer

Phase 1 — Discover

AI endpoint discovery maps all LLM interfaces, API surfaces, MCP server configurations, RAG data sources, agent tool chains, and model deployment architectures. Technology fingerprinting identifies underlying models, frameworks, and guardrail implementations.

Phase 2 — Analyze

NIST AI RMF-aligned risk assessment evaluates governance, data pipeline security, model training procedures, and deployment controls. Threat models map AI-specific attack vectors including prompt injection, data poisoning, and tool-use chain exploitation.

Phase 3 — Attack

LLM Attack Suite deploys 1,005+ adversarial prompts across all 10 OWASP LLM categories. MCP tool injection and privilege escalation testing. Multi-modal adversarial attacks across text, image, and API inputs. Exhaustive escalation until bypass or confirmed resilience.

Phase 4 — Report

AI-specific technical reports with LLM vulnerability categorization (OWASP LLM Top 10), MCP security findings, adversarial red team results with bypass evidence, and NIST AI RMF compliance gap analysis for governance teams.

Phase 5 — Remediate

AI-specific remediation — guardrail configuration, input validation rules, output filtering policies, prompt engineering improvements, MCP authentication hardening, and model monitoring setup. Retest validates every control.

DELIVERABLES

What You Receive

Every engagement produces four comprehensive deliverables designed for different stakeholders — from technical teams requiring reproduction steps to executives needing business impact context.

Technical Report	Management Report	Threat Model Report	ToIP Playbook
CVSS-scored, CWE-mapped findings with proof-of-concept evidence and step-by-step reproduction instructions for every vulnerability discovered.	Executive-level risk summary with business impact analysis, prioritized recommendations, and strategic remediation roadmap for leadership.	Comprehensive threat landscape analysis with attack vectors mapped to MITRE ATT&CK framework specific to your environment.	Target-specific attack playbooks generated from 15,000+ real-world disclosed vulnerabilities via our RAG-powered intelligence engine.

- ✓ **Technical Report**
  Detailed findings with CVSS scoring, CWE mapping, proof-of-concept evidence, and step-by-step reproduction instructions.
- ✓ **Management Report**
  Executive-level risk summary with business impact analysis, prioritized recommendations, and strategic remediation roadmap.
- ✓ **Threat Model Report**
  Comprehensive threat landscape analysis specific to your environment, attack vectors, and adversary profiles mapped to MITRE ATT&CK.
- ✓ **ToIP Playbook Document**
  Target-specific attack playbooks generated from 15,000+ real-world disclosed vulnerabilities via our RAG-powered intelligence engine.

All deliverables are produced within the agreed timeline with milestone-driven tracking. Critical findings are escalated immediately via the OffSec Management Portal — you never wait for the final report to learn about urgent risks.

Ready to Secure Your Organization?

Partner with Technieum for AI-powered offensive security with zero false positives.

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